Code Connections

OOP & Database

What is OOP

- A programming paradigm that organizes data into objects
- Objects contain **attributes** (data) and **methods** (functions)
- Helps represent real-world entities like books, cars, or users in code

Key OOP Concepts

- **Class:** Blueprint for creating objects (e.g., Car, Book, Weather)
- **Object:** An instance of a class (e.g., a specific book or a car model)
- **Attributes:** Properties of an object (e.g., title, author, publication year)
- **Methods:** Actions an object can perform (e.g., borrowing a book)

Code Structure for OOP Implementation

1. **Book Class:** Handles fetching data from the API and parsing it
2. **Database Class (SQLAlchemy):** Manages data storage and retrieval
3. **Main File (main.py):** Serves as the entry point to integrate everything

Introduction to SQLAlchemy (Database Class)

- **What is SQLAlchemy?**
  - An Object-Relational Mapper (ORM) for Python
  - Allows interaction with databases using Python objects instead of SQL
- **Setting Up SQLAlchemy:**
  - Define database models as Python classes

Steps to Connect to a DB

- define the table structure
- Initialize the database connection
- Create a session factory
- Implement CRUD: create, read, update, delete

Summary

- Defining the Table
- Creating a Connection
- Managing Sessions
- Interacting with the Database

Creating the Main

- Fetch the book data from an external API
- Fetch and parse the data
- Insert fetched books into the database
- Retrieve and display books from the database
- Update a book
- Delete a book
- Retrieve books again to confirm changes

Resources

- Full Open Library API documentation: https://openlibrary.org/developers/api
- DB Browser for SQLite: https://sqlitebrowser.org/